

1 Question (16 points)

Design a single tape TM that recognizes the language $L = \{0^n 1^{n^3} | n \geq 1\}$. Do not describe the steps of TM. Draw a state diagram, give states, transitions, etc.

2 Question (17 points)

Let $B_{(m,n)} = \{x \mid x \text{ is a number in modulus } m \text{ that is a multiple of } n\}$. Show that for each $m, n \geq 1$, the language is $B_{(m,n)}$ is regular.

3 Question (17 points)

Give PDA that generates the following language. The alphabet is $\{0, 1\}$.

$D = \{w \mid w \text{ contains at least two 0s and two 1}\}$